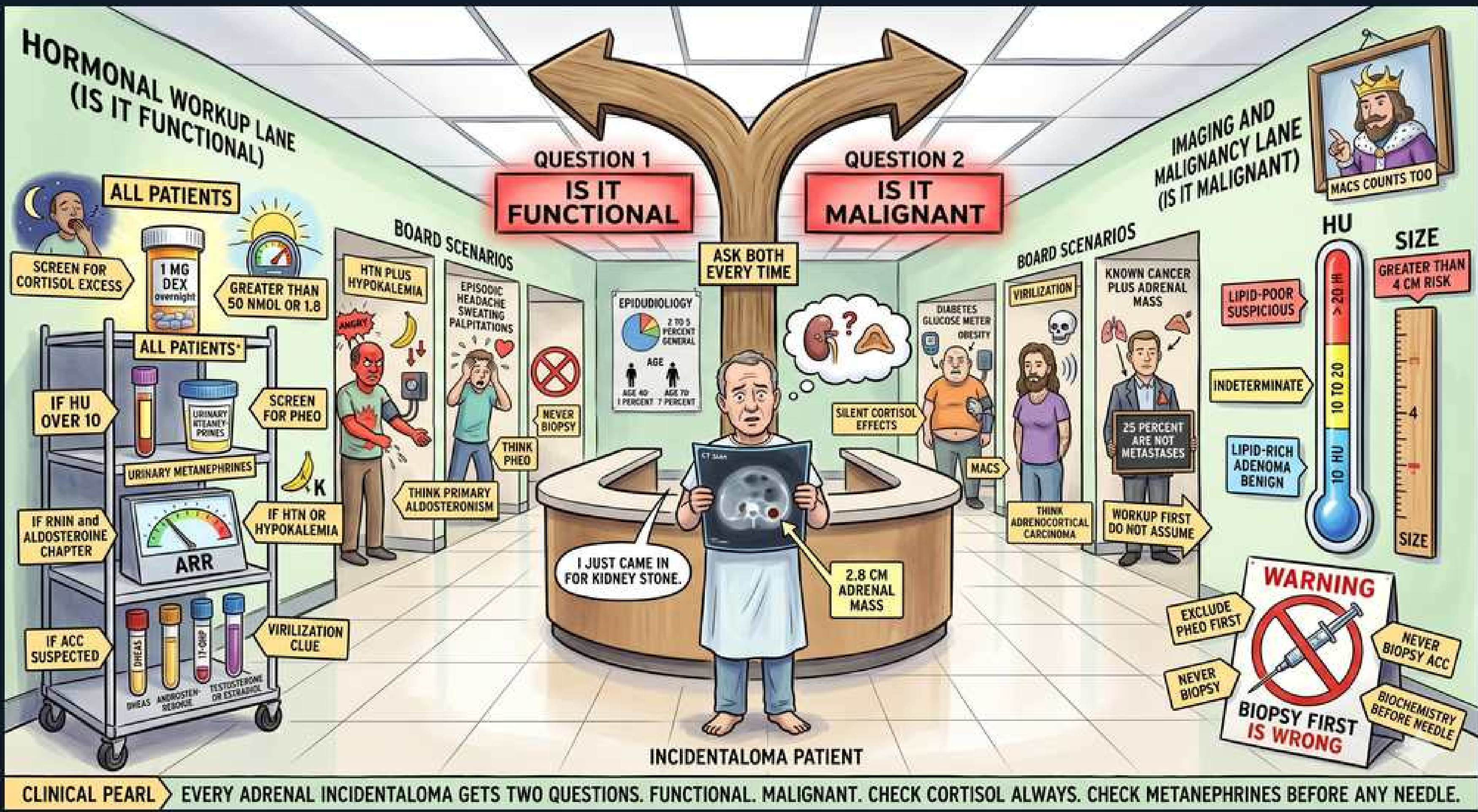
Adrenal Incidentaloma — Two Questions: Functional & Malignant



1. Ask both questions every time

WHAT YOU SEE

Triage fork: Question 1 functional, Question 2 malignant

WHAT IT ENCODES

Every adrenal incidentaloma gets TWO mandatory questions: (1) Is it functional (hormone-secreting)? (2) Is it malignant? Both must be answered regardless of how it was found.

BOARD TRAP

Skipping the hormonal workup because the mass looks benign on CT — both questions are always required.

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2. Screen cortisol in ALL

WHAT YOU SEE

All patients: 1 mg overnight dexamethasone suppression test

WHAT IT ENCODES

Every incidentaloma is screened for autonomous cortisol with a 1 mg overnight dexamethasone suppression test. Failure to suppress (cortisol >1.8 μg/dL) indicates subclinical/mild autonomous cortisol secretion.

BOARD TRAP

1 mg DST is the universal screen — cortisol >1.8 after dex = abnormal (cannot suppress).

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3. Screen metanephrines before biopsy

WHAT YOU SEE

All patients: urine/plasma metanephrines, clue to virilization

WHAT IT ENCODES

Plasma or 24h urine fractionated metanephrines are checked in every incidentaloma to exclude pheochromocytoma BEFORE any biopsy or surgery, since unrecognized pheo can trigger fatal hypertensive crisis.

BOARD TRAP

Always exclude pheo with metanephrines before biopsy — never biopsy first.

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4. ARR if hypertensive/hypokalemic

WHAT YOU SEE

ARR test if HTN or hypokalemia

WHAT IT ENCODES

Add aldosterone-to-renin ratio only if the patient is hypertensive or hypokalemic, screening for primary aldosteronism. A high ARR with autonomous aldosterone confirms Conn syndrome.

BOARD TRAP

ARR is selective (HTN/hypokalemia), unlike the universal cortisol and metanephrine screens.

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5. Never biopsy first

WHAT YOU SEE

Center: 'I just came in for a kidney stone, 2.8 cm adrenal mass'

WHAT IT ENCODES

The incidentaloma is found while imaging for something else. The workhorse rule is biochemical screening before any tissue sampling — biopsy is almost never the first step.

BOARD TRAP

Biopsy first is wrong — hormone screen (especially metanephrines) must precede sampling.

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6. Functional board scenarios

WHAT YOU SEE

Board scenarios for 'is it functional': Cushingoid, HTN, palpitations

WHAT IT ENCODES

Functional clues: episodic headache/sweating/palpitations → pheo; resistant HTN + hypokalemia → aldosteronoma; central obesity/striae/easy bruising → cortisol-secreting adenoma. Each maps to a specific screening test.

BOARD TRAP

Match the clinical clue to the right hormone test; an asymptomatic mass can still be biochemically functional.

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7. Silent cortisol effects

WHAT YOU SEE

Diabetes/osteoporosis icons: silent cortisol effects

WHAT IT ENCODES

Mild autonomous cortisol secretion is often clinically silent yet drives diabetes, hypertension, and osteoporosis. This is why the 1 mg DST is done even without overt Cushing features.

BOARD TRAP

'No Cushingoid look' does not exclude cortisol excess — subclinical secretion still harms; screen everyone.

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8. Malignancy: known cancer = met

WHAT YOU SEE

Board scenario: known cancer plus adrenal mass

WHAT IT ENCODES

In a patient with a known primary malignancy, a new adrenal mass is a metastasis until proven otherwise (lung, breast, melanoma, renal commonly metastasize to adrenals).

BOARD TRAP

In active cancer, assume metastasis — but still exclude pheo biochemically before any biopsy.

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9. ACC suspicion (large mass)

WHAT YOU SEE

Board scenario: think adrenocortical carcinoma

WHAT IT ENCODES

Adrenocortical carcinoma is suggested by large size (>4 cm), rapid growth, irregular margins, high unenhanced HU, and mixed hormone hypersecretion. ACC is aggressive with poor prognosis.

BOARD TRAP

Large/heterogeneous masses raise ACC concern — size and imaging phenotype drive the malignancy call, not biopsy.

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10. HU < 10 = benign adenoma

WHAT YOU SEE

Hounsfield gauge: <10 HU lipid-rich, adenoma benign

WHAT IT ENCODES

On unenhanced CT, ≤10 HU indicates a lipid-rich benign adenoma. Indeterminate density (>10 HU) needs washout CT or further characterization. HU is the key imaging discriminator.

BOARD TRAP

Lipid-rich (≤10 HU) = benign adenoma; higher HU is indeterminate and needs washout, not automatic surgery.

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11. Size > 4 cm raises resection

WHAT YOU SEE

Size thermometer: >4 cm risk

WHAT IT ENCODES

Masses larger than 4 cm carry higher malignancy (ACC) risk and generally warrant surgical resection (adrenalectomy), even if biochemically silent.

BOARD TRAP

>4 cm or growing → resect; small (<4 cm), low-HU, non-functional masses can be surveilled.

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12. Never biopsy without excluding pheo

WHAT YOU SEE

Red warning: exclude pheo, biopsy first is wrong

WHAT IT ENCODES

Biopsy of an adrenal mass is reserved for suspected metastasis/lymphoma AFTER pheochromocytoma is biochemically excluded. Biopsying a pheo can cause lethal catecholamine crisis.

BOARD TRAP

Single biggest incidentaloma trap: biopsying before excluding pheo with metanephrines.

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13. Clinical pearl banner

WHAT YOU SEE

Banner: every incidentaloma gets two questions, check cortisol and metanephrines

WHAT IT ENCODES

Pearl: every adrenal incidentaloma = ask functional? and malignant? Always check cortisol (1 mg DST) and metanephrines; add ARR if hypertensive/hypokalemic; image with unenhanced HU and size.

BOARD TRAP

The scene reduces to: dual workup, universal cortisol + metanephrine screen, no biopsy before pheo exclusion.

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14. VMA/metanephrine before any needle

WHAT YOU SEE

Lower-left urinary metanephrine sample

WHAT IT ENCODES

24-hour urinary or plasma free metanephrines/normetanephrines are the catecholamine screen; positive results mandate alpha-blockade before any surgery, never beta-blockade first.

BOARD TRAP

Start alpha-blockade (phenoxybenzamine) before beta-blocker in confirmed pheo — unopposed alpha causes crisis.
